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Element 010 Assignment

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Table of Contents

Table of Figures 3
Part 1 4
Part 2 10
Abstract 10
1 Introduction 10
2 Mobile Operating System Security Issues 11
2.1 Malware 11
2.2 History 12
3 Android Security Challenges 13
4 Countermeasures 13
5 Conclusion 14
References 15

Table of Figures

Figure 1 - Two-level Page Table Logical Address 8
Figure 2 - Two-Level Page Table Address Translation 8
Figure 3 - The Five-State Process Lifecycle (Subham Datta, 2022) 9
Figure 4 - Worldwide Market Share of Mobile Operating Systems 11
Figure 5 - Screenshot of Movie Player application 12

Part 2

A Report on the Current Challenges of Security of Operating Systems of Mobile Devices

Abstract

Mobile devices are now being used as targets for attacks by threat actors. They (mobile devices) are not exempted from malicious attacks, such as malware infections. In fact, the development of new mobile devices equates to new security issues. This paper discusses the security challenges of mobile devices' operating systems, with a focus on the Android Operating System, while also proffering security measures to be put in place.

Keywords – Mobile device, Android, Malware

1 Introduction

Many years ago, people did not need to think about security issues, especially in relation to mobile devices. The types of mobile devices that were available then were specifically for making calls. Now, the world is advancing, innovation is growing exponentially, and new technologies (including mobile devices) are evolving. The advent of these technologies has called for the need for security awareness.

(Viswanathan Priya, 2022) described a mobile device as the name for a computer held in the hand. Having a mobile device has become fundamental, even if not for personal use. For organizations, the use of mobile devices by employees to connect to company emails, respond to urgent requests, and attend meetings has come to stay (Gosselin Susan, 2022). These mobile devices are built with different operating systems, some of which are Android, iOS, Blackberry and Windows. Research has shown that the Android OS tops the list in most cases. Figure 2 (Statcounter, 2022) shows the different mobile operating systems and their percentage market share in the world between January 2021 and November 2022.

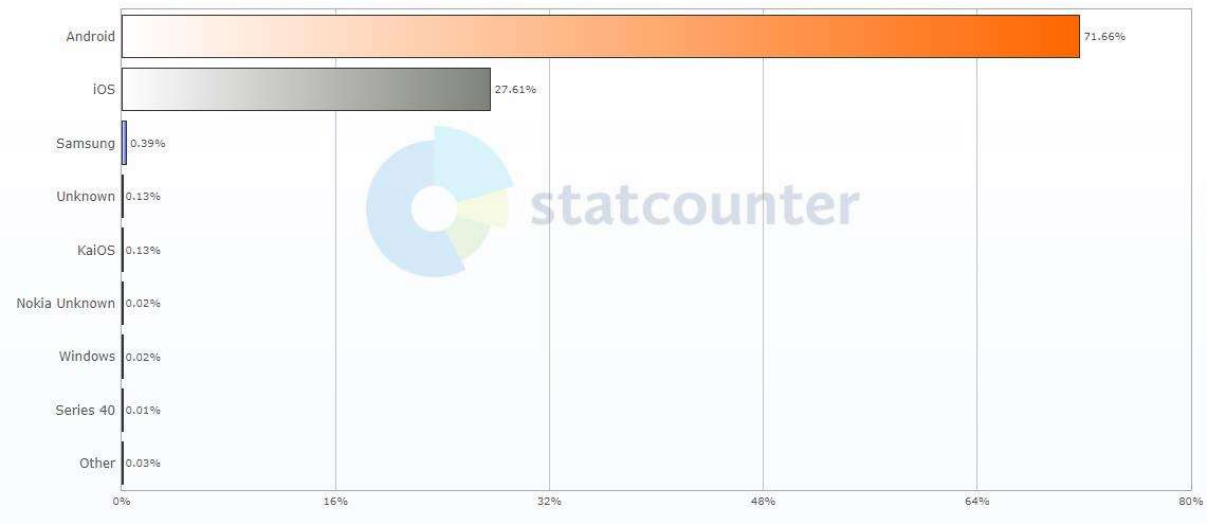

Figure 4 - Worldwide Market Share of Mobile Operating Systems

The remaining sections of this paper will discuss the security issues in these operating systems, with a focus on the Android OS and the measures to put in place to handle these issues.

2 Mobile Operating System Security Issues

Most mobile devices are always connected to the Internet, making them susceptible to attacks. Apart from being connected to the Internet, they can also connect to other devices and share resources. For instance, a mobile device can connect to a computer via USB or Bluetooth to share some data. The way personal computers are prone to cyber threats is now the same way mobile devices are exposed to different cyber threats. Dealing with security challenges requires an understanding of some security concepts (Ting Zhao, Gang Zhang and Lei Zhang, Dec 2014).

2.1 Malware

Malicious software (also known as malware) is a type of software that is designed with the intent to harm a system or network (Lutkevich Ben, 2022). Malware exists in different forms, and it can be classified based on some specifications. It can be how it is distributed, performed action on the infected machine, or its functionality. The common types of malware are listed below.

a. Virus: A malicious executable code that is attached to another program (or document), which is designed to spread from one device to another.

b. Worms: Just like the virus, worms are also designed to spread from one device to another. However, they do not require the input of the host to infect the system. It replicates itself over and over to decrease the device resources (memory and bandwidth), thereby causing the slowness of the device.

c. Trojan Horse: This type of malware disguises itself as a legitimate program. It uses social engineering on unsuspecting users to download 'legitimate' software. When downloaded, it creates a backdoor on the system and steals sensitive data (Fortinet, 2022).

d. Adware: Adware, also known as advertisement-supported software, pushes unwanted pop-up advertisements at users when using the browser. These adverts are usually not related to the sites they are visiting. The main purpose of adware is financial gain.

e. Spyware: Malicious software installed on a device without the user's consent is spyware. It gets into the device through phishing emails or when a user downloads software from an unreliable site. It runs in the background, monitors activities done on the device, collects user and device information, and sends it to third parties who will use the information for financial gain.

f. Ransomware: Ransomware is malicious software designed to take control of one's device or data by locking them, in exchange for money (ransom). It encrypts the files so that the user cannot access them, and requests money to decrypt or release the files. Even when the money is paid, there may be no assurance that the user will have control of the file.

2.2 History

In 2010, an employee of Kaspersky, (Maslennikov Denis, 2010) reported the first Android OS malware, FakePlayer. This malware is a Trojan SMS that poses as a media player application and prompts users to install it. When installed by the user, it creates an icon "Movie Player" (see Figure 3) which is displayed alongside other applications. It sends SMS to prime rate numbers outside the user's consent, each costing about $5. This results in financial loss to the user.

Figure 5 - Screenshot of Movie Player application

In 2016, another malware, HummingBad was discovered (Check Point, 2016). This malware displayed malicious advertisements, mimics clicks on the advertisements and installs

deceptive applications on the devices. As with most malware attacks, the aim of this malware attack is financial gain (Woollaston-Webber Victoria, 2016).

3 Android Security Challenges

Android was designed to be an open OS built on the Linux kernel. It allows several applications to run concurrently. Being an open source means the source code is available to the public. A user is likely to install some of the applications from unidentified sources, instead of getting the applications from the Google Play Store.

(Chatterjee et al., 2016) discussed that android's Google Play store is one of the causes of possible security hazards. Because Google Play is not a controlled environment, the chance that applications there may be embedded with malware is high. Users who download applications are oblivious to the degree of access applications can have on their phones. When this vulnerability is exploited by cybercriminals, it can lead to data loss and performance issues.

In another research by (Ting Zhao, Gang Zhang and Lei Zhang, Dec 2014), the attacks on Android mobile devices were classified into four types. The hardware-based attack focuses on the hardware the OS is running on. Any attack on this hardware will bring about terrific damage, if successful. The Linux kernel-based attack occurs when the kernel security options are disabled. Certifying the authenticity of user-installed applications is usually not possible in most cases. This makes application attacks the simplest method of attack. The core program attack is in relation to the Dalvik process. Malicious codes can be executed if there is an error in the Dalvik process when calling the system function.

4 Countermeasures

The Merriam-Webster dictionary defines countermeasure as an action designed to negate another action. (Chatterjee *et al.*, 2016) suggested the use of cryptography to encrypt the data on the device on which the Android OS is installed. With this, access to personal information and the loss of data to threat actors will be reduced.

Also, the signing of applications with a digital certificate should be made mandatory by Android. This was corroborated by (Delac, Silic and Krolo, May 2011). This is to secure the application in order not to be used for malicious acts, as in the case of FakePlayer. The certificate will be used to authenticate the application owner.

Antimalware should be installed on mobile devices to prevent malware infection. Most antivirus/antimalware vendors now have solutions that can protect mobile devices.

5 Conclusion

Mobile devices have become part of our lives. A successful attack on one will lead to the leakage of private data. Android OS is widely used because it is open to the public, which makes it vulnerable to exploits. Security challenges in Android OS were discussed, with security measures to be put in place. Mobile devices should be secured with the same degree of protection given to computers.

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